



- 6 (a) Compare an α -particle with a β^+ particle in terms of their masses and charges.

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..... [3]

- (b) Nucleus P undergoes α -decay to form nucleus Q. Nucleus Q then undergoes a further decay to form nucleus R. The proton and nucleon numbers of P and R are shown in Fig. 6.1.

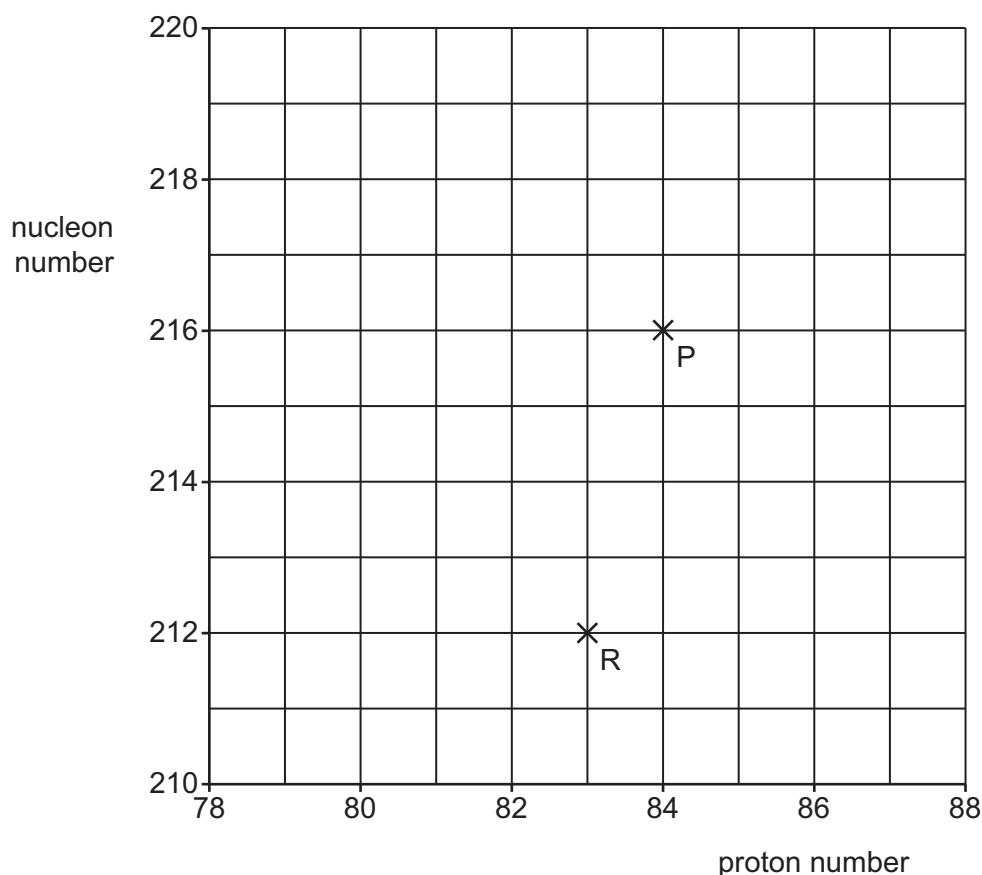


Fig. 6.1

- (i) On Fig. 6.1, draw a cross (x) to show the proton number and nucleon number of Q. Label your cross Q. [1]
- (ii) State the names of the particles emitted as Q decays to form R.

.....

..... [2]



- (c) Before the α -decay, P is travelling at a constant velocity. After the decay, Q has a velocity of $1.3 \times 10^5 \text{ m s}^{-1}$ at an angle of 68° to the original path of P. The α -particle has a velocity of $150 \times 10^5 \text{ m s}^{-1}$ at an angle of θ to the original path of P, as shown in Fig. 6.2.

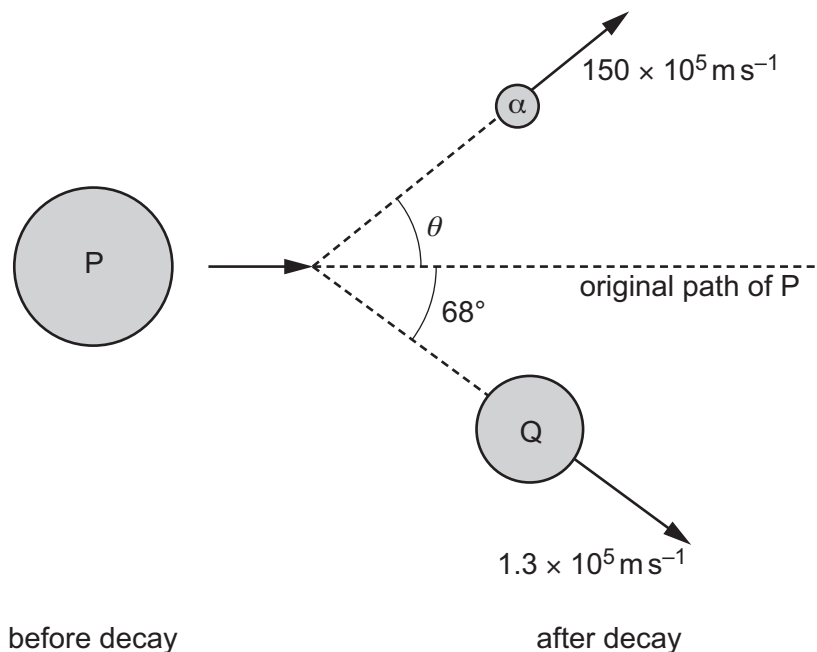


Fig. 6.2 (not to scale)

- (i) Use the principle of conservation of momentum to determine θ .

$$\theta = \dots\dots\dots^\circ \quad [3]$$

- (ii) Calculate the kinetic energy of the α -particle.

$$\text{kinetic energy} = \dots\dots\dots \text{ J} \quad [2]$$

[Total: 11]





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